

Phase 4

Dosage Sensitivity, Perturbation, Functional Convergence, and Integrated Adjudication

Abstract

This frozen Phase 4 master report consolidates Phases 4A through 4G of the project "Genomic and Molecular Analysis of a 17p13.3 Microdeletion and Its Candidate Visual-Neurological Mechanisms." The structural reference remains GRCh37/hg19 chr17:1,411,408-1,841,103, a nominal 429,696-bp heterozygous deletion affecting at least seventeen annotated loci, including thirteen protein-coding genes. Phase 4 tested whether heterozygous dosage reduction can move from genomic plausibility into experimentally observable molecular, cellular, developmental, visual-system, and seizure-relevant consequences. The phase explicitly separated four genetic states - heterozygous loss, biallelic or complete loss, hypomorphic reduction, and specific pathogenic variants - and treated causality as the final rather than initial inference.

Across the complete audit, the strongest evidence supports a multi-layer functionalreserve model rather than a single deleted "photosensitivity gene." RPA1 provides the clearest direct human haploinsufficiency phenotype, SERPINF1 provides direct heterozygous dosage reduction of PEDF, INPP5K provides one-copy physiological evidence outside the retina and strong phosphoinositide/lysosomal mechanism, and several other genes produce relevant phenotypes under severe or variant-specific perturbation. PRPF8, WDR81, SERPINF1/PEDFassociated biology, and related systems converge on photoreceptor integrity; PITPNA, WDR81, RILP, and INPP5K converge on membrane identity and intracellular material handling; SERPINF1, RPA1, and SCARF1 define a layered protection-repair-clearance architecture; and WDR81, INPP5K, RTN4RL1, and RILP converge on neuronal development

and circuit organization. These convergences are biologically meaningful but do not constitute demonstrated synergy or epistasis.

The retinal evidence favors reduced resilience, impaired maintenance, delayed recovery, and stress vulnerability substantially more strongly than increased photon-response gain. The strongest partial-dose retinal physiology identified during Phase 4 shows reduced electroretinographic output, not amplified output. Complete RTN4RL1 loss in human neurons alters network construction but decreases synapse density and synchronous calcium-network activity, arguing against a simple generalized-hyperexcitability interpretation. MIR22 provides the strongest seizure-relevant locus-level evidence, but the decisive chronic epilepsy phenotype was shown after complete loss and a prior epileptogenic insult, not after heterozygous loss or photic stimulation.

Phase 4F localized the central causal gap. Human photosensitive epilepsy depends on

frequency-sensitive visual-cortical responses, inhibitory containment, and abnormal large-scale network propagation. The deletion has substantial biology at the retinal end and meaningful general neuronal/white-matter biology downstream, but no current experiment demonstrates abnormal lateral geniculate physiology, V1 flicker responses, visual-cortical inhibition, 10-20-Hz entrainment, pathological occipital-to-distributed-network propagation, a photoparoxysmal response, or visually triggered seizures produced by this exact CNV. Phase 4G therefore classifies the deletion as a biologically consequential, mechanistically plausible candidate modifier of visual-neural and epileptogenic reserve, not an established cause of epileptic photosensitivity.

Phase 4 is frozen as Version 1.0 as of 15 September 2026. The frozen interpretation may

be superseded only by explicitly reopened analysis or by new perturbational, experimental, or clinical evidence. Phase 5 is reserved for perturbational validation against independent in silico or existing experimental datasets.

Precedence

1. Structural and Evidentiary Precedence

The structural coordinates used throughout Phase 4 are the original GRCh37/hg19 coordinates chr17:1,411,408-1,841,103. GRCh38 and current annotations may be used to reconcile transcripts or expression resources, but they do not replace the primary CNV definition. The interval includes thirteen protein-coding genes - INPP5K, PITPNA, SLC43A2, SCARF1, RILP, PRPF8, TLCD2, WDR81, SERPINF2, SERPINF1, SMYD4, RPA1, and RTN4RL1 - and four noncoding loci - PITPNA-AS1, MIR22HG, MIR22, and RN7SL105P. INPP5K and RTN4RL1 are boundary genes. The reported-coordinate model removes substantial INPP5K coding sequence and approximately 99 percent of the canonical RTN4RL1 coding sequence on the affected allele, illustrating why genomic overlap percentage cannot be equated with coding disruption percentage.

The controlling evidence hierarchy for Phase 4 was genomic evidence -> molecular mechanism -> expression/spatiotemporal opportunity -> perturbational evidence -> cellular phenotype -> developmental consequence -> visual-system physiology -> phenotype association -> causality. This hierarchy was used because rare-CNV interpretation is especially vulnerable to overreading knockout phenotypes, assuming that one-copy loss produces exactly 50 percent protein function, or transferring a specific dominant-negative or gain-of-function disease mechanism to a simple deletion.

2. Dosage-Sensitivity Precedence

RPA1 provides the strongest direct precedent that a one-copy genomic loss inside this broader chromosomal region can produce a measurable human molecular and cellular phenotype. O'Driscoll and colleagues studied contiguous-gene deletion cell lines in which RPA1 was haploinsufficient and found reduced RPA1 abundance, impaired ATR-dependent DNA-damage signaling, defective checkpoint behavior after ultraviolet exposure, and increased stress-associated nuclear fragmentation. Restoration of RPA1 rescued key abnormalities, supporting a gene-specific haploinsufficiency effect rather than an unspecified contiguous-deletion artifact (O'Driscoll et al.).

SERPINF1 provides a second clinically relevant dosage precedent. Heterozygous carriers of truncating SERPINF1 variants had reduced SERPINF1 transcript levels and significantly lower serum PEDF concentrations without the severe skeletal phenotype of biallelic SERPINF1 disease (Al-Jallad et al.). This establishes that one functional copy can measurably lower the gene product even when a classic Mendelian phenotype is absent.

INPP5K heterozygous mouse data show that one-copy reduction can alter physiological signaling outside the retina. Ijuin and colleagues found increased insulin sensitivity and altered Akt-dependent glucose uptake in heterozygous SKIP/Inpp5k-deficient mice, demonstrating that a one-copy state is not necessarily molecularly silent (Ijuin et al.). By contrast, WDR81 and several other loci show dramatic phenotypes primarily under biallelic or severe loss, while PITPNA displays substantial developmental buffering through the closely related paralog PITPNB. These contrasting precedents required gene-specific rather than uniform dosage assumptions.

SERPINF2 serves as an important methodological positive control: heterozygous alpha2antiplasmin deficiency can produce a clear biochemical dosage phenotype while remaining mechanistically remote from retinal photon detection. Phase 4 therefore treated "dosagesensitive" and "photosensitivity-relevant" as separate axes rather than synonyms.

3. Molecular and Chemical Precedence

3.1 TLCD2 and Membrane Composition

Ruiz and colleagues showed that human TLCD1/2 influence membrane phospholipid fatty-acyl composition and physical fluidity. TLCD2 knockdown altered incorporation of longchain polyunsaturated fatty acids into phosphatidylcholine and phosphatidylethanolamine and protected cells from palmitate-induced membrane rigidification under the tested conditions (Ruiz et al.). The result establishes a real membrane-composition phenotype but also supplies counterevidence against a simple assumption that TLCD2 loss is intrinsically harmful.

Petkevicius and colleagues later showed tissue- and paralog-dependent phosphatidylethanolamine effects in *Tlcd*-family mouse models, emphasizing context and redundancy.

3.2 SERPINF1/PEDF and PNPLA2/PEDF-R

PEDF is a retinally important secreted protein encoded by SERPINF1. Subramanian and colleagues identified functional PEDF-R/PNPLA2 interactions and linked PEDF signaling to phospholipase/lipase activity and retinal cell survival (Subramanian et al.). Bernardo-Colon and colleagues subsequently showed that *Pnpla2* ablation causes photoreceptor degeneration, altered retinal proteins and lipid species, apoptosis, and attenuated retinal electrophysiology; heterozygous animals already demonstrated measurable physiological impairment (Bernardo-Colon et al.). Rebutini and colleagues showed that *Serpinf1* deletion produces senescence-like

RPE changes, reduced Pnpla2/PEDF-R expression, and accumulation of rhodopsin and lipids after outer-segment phagocytosis (Rebustini et al.). Hara and colleagues further demonstrated that PNPLA2 mobilizes retinyl esters from RPE retinosomes and that loss delays 11-cis-retinal regeneration and dark adaptation (Hara et al.).

The PEDF-DHA literature supplies an additional stress-protection precedent. Mukherjee, Marcheselli, and Bazan demonstrated production of the DHA-derived mediator neuroprotectin D1 in human RPE cells and showed protection against oxidative-stress-induced apoptosis (Mukherjee et al.). This supports a chemically plausible SERPINF1/PEDF-linked retinal stressdefense layer without establishing that heterozygous SERPINF1 loss alone reproduces the complete downstream phenotype.

3.3 INPP5K

McGrath and colleagues demonstrated that INPP5K hydrolyzes PI(4,5)P₂ to PI4P on autolysosomal membranes and is required for autophagic lysosome reformation. Strong loss produces PI(4,5)P₂ accumulation, PI4P depletion, persistent clathrin-associated reformation tubules, enlarged autolysosomes, lysosome depletion, and impaired autophagic homeostasis; wild-type INPP5K or reduction of PI(4,5)P₂ synthesis can rescue key defects (McGrath et al.). The evidence is mechanistically strong but is principally derived from skeletal-muscle systems rather than retina.

3.4 WDR81

Liu and colleagues identified WDR81/WDR91 as negative regulators of endosomal PI3P and early-to-late endosome conversion. WDR81/WDR91 loss increases Beclin1/VPS34-associated activity, elevates PI3P-rich early-endosomal identity, enlarges endosomes, and impairs receptor degradation (Liu et al.). This places WDR81 upstream in a

membrane-identity and cargo-maturation system distinct from INPP5K's autolysosomal PI(4,5)P₂/PI4P function.

3.5 RILP

RILP is a Rab7A effector and dynein-associated adaptor rather than an enzyme. Yap and colleagues demonstrated that disruption of the RAB7A-RILP interaction decreases retrograde late-endosome transport in dendrites, impairs dendritic cargo clearance, and reduces dendritic arborization even when somatic lysosomal degradation remains comparatively preserved (Yap et al.). This distinguishes physical cargo transport from terminal degradation and supports a mechanochemical rather than catalytic interpretation of RILP.

3.6 PRPF8

PRPF8 is a core U5 spliceosomal scaffold, not the catalytic metal-binding enzyme or ATPase of the spliceosome. Human retinal-organoid models carrying pathogenic heterozygous PRPF8 variants show that retina-specific transcript processing is unusually vulnerable to PRPF8 dysfunction. The H2309P model demonstrates altered splice-site selection, ciliary/polarity abnormalities, and photoreceptor loss, while the Y2334N model preserves early retinal specification but later develops markedly reduced photoreceptor inner/outer-segment structures and retinal splicing abnormalities (Zimmann et al.; Atkinson et al.). These findings strongly establish PRPF8 retinal relevance but do not demonstrate that whole-allele haploinsufficiency behaves like the pathogenic mutant proteins.

3.7 RPA1 and Photochemical Stress

The RPA1 precedent becomes relevant to light injury because retinal cells can experience photochemically generated DNA damage. Sparrow, Zhou, and Cai showed that A2E-containing RPE exposed to 430-nm illumination accumulates oxidative DNA lesions, including 8-oxo-dG,

and that a singlet-oxygen quencher reduces the damage (Sparrow et al.). Combining this independent retinal-stress literature with direct RPA1 haploinsufficiency establishes a credible but still untested retinal-reserve hypothesis: the same light-associated lesion burden could become harder to manage when DNA-damage-response reserve is reduced.

3.8 Secondary Systems

SCARF1 mediates C1q-associated recognition and efferocytosis of apoptotic cells; SCARF1 deficiency impairs clearance and allows apoptotic and secondary-necrotic material to accumulate (Ramirez-Ortiz et al.). SLC43A2/LAT4 is a facilitated-diffusion transporter for neutral essential amino acids, with global or tissue-specific deletion producing strong systemic transport phenotypes but limited direct retinal evidence (Guetg et al.; Rajendran et al.). SMYD4 is a SET/MYND-domain methyltransferase with experimentally demonstrated developmental and chromatin effects, including H3K4-associated regulation and HDAC1 interaction in zebrafish cardiac development, but no established visual-neural haploinsufficient phenotype (Xiao et al.). These systems were therefore retained as secondary reserve mechanisms rather than core photosensitivity drivers.

4. Cellular-Phenotype Precedence

Several constituent-gene perturbations cross the threshold from molecular mechanism into bona fide retinal or neuronal cellular phenotypes. Pathogenic PRPF8 variants produce human retinal-organoid photoreceptor and outer-segment abnormalities. Severe WDR81 dysfunction causes progressive photoreceptor loss in mice, and wild-type WDR81 rescue restores survival. The PEDF-R/PNPLA2 system produces photoreceptor deformation, apoptosis, altered retinal proteins, and reduced ERG output. These independent systems therefore converge on photoreceptor integrity through different upstream mechanisms.

At the RPE level, complete *Serpinf1* loss causes senescence-associated changes, impaired outer-segment processing, and rhodopsin/lipid accumulation, while heterozygous *Serpinf1* loss already reduces downstream *Pnpla2* expression. *PNPLA2* loss further demonstrates delayed retinoid mobilization and dark adaptation. The source literature thus supports genuine RPE support-system dysfunction but does not establish that the complete heterozygous CNV itself has been tested under visible-light challenge.

For neural cells, *RTN4RL1* is especially informative. Wang and colleagues showed that complete loss in human neurons increases axonal and dendritic elaboration but decreases synapse formation, multiple synaptic proteins, and synchronous calcium-network activity (Wang et al.). This provides strong evidence that the gene can reorganize developing human neuronal networks while simultaneously arguing against a simplistic hyperexcitability model.

5. Developmental Precedence

WDR81 provides the strongest early corticogenesis precedent. Carpentieri and colleagues showed that *WDR81* loss alters endosomal EGFR trafficking, reduces MAPK signaling, slows radial-glial proliferation during mid-to-late neurogenesis, and ultimately reduces brain size. A Cyclin D2 manipulation rescues the proliferation phenotype, providing mechanistic continuity from endosomal trafficking to cell-cycle control (Carpentieri et al.). The experiments principally involve strong or complete loss, so transfer to heterozygous CNV dosage remains unresolved.

PITPNA/PITPNB studies provide an explicit compensation precedent: severe combined loss disrupts cortical progenitor polarity, Golgi phosphoinositide signaling, interkinetic nuclear migration, and cortical development, whereas one remaining functional copy of either paralog can support grossly normal brain development. This demonstrates that pathway essentiality does not imply single-gene haploinsufficiency.

RTN4RL1 and RILP occupy later developmental windows. RTN4RL1 loss alters neurite growth and synaptogenesis in human neurons; RILP-dependent transport is required for normal dendritic arborization. The two effects can even point in opposite morphological directions, establishing topological convergence on circuit construction without directional convergence. SERPINF1/PEDF contributes to postnatal retinal maturation and vascular/stress biology, while PRPF8 pathogenic variants show preserved early retinal specification followed by later photoreceptor maturation failure. MIR22 differs again: complete loss leaves baseline brain structure comparatively normal but worsens chronic epilepsy after an epileptogenic insult, placing its strongest effect in activity-dependent remodeling rather than constitutive brain formation (Silva et al.).

6. Human CNV and White-Matter Precedence

The most directly relevant clinical CNV precedent comes from interstitial 17p13.3 deletions located between YWHAE/CRK and PAFAH1B1. Emrick and colleagues reported multifocal white-matter abnormalities in five of six individuals and narrowed a smallest overlapping region to approximately 345 kb containing eleven RefSeq genes, with RTN4RL1 proposed as the strongest candidate (Emrick et al.). This evidence is substantially closer to the present interval than classic 17p13.3 deletion syndromes involving PAFAH1B1 or YWHAE/CRK. It supports neurological consequence and incomplete penetrance but does not provide a replicated photoparoxysmal or visually triggered seizure phenotype.

7. Photosensitive-Epilepsy Precedence

The disease physiology sets a high bar for any proposed genetic mechanism. Photosensitive epilepsy is not simply excessive retinal photon detection. Timar and colleagues found that PSE patients show increased occipital PPR power particularly during 10-20-Hz

stimulation, while ordinary transient and steady-state visual-evoked responses are not uniformly elevated. Propagating PPRs are associated especially with abnormal occipitofrontocentral connectivity (Timar et al.). Benedetto and colleagues further reported greater visually evoked EEG energy but reduced phase coherence and unstable alpha dynamics in photosensitive idiopathic generalized epilepsy (Benedetto et al.). Earlier MEG work found gamma-band phase clustering before transitions into PPRs (Parra et al.).

Pattern-sensitive epilepsy further localizes the critical transformation to organized cortical visual processing. Spatial frequency, orientation, contrast, and stimulus extent alter epileptogenicity, and classic work argued against a trigger confined to the LGN (Soso, Lettich, and Belagum; "Neurophysiological Aspects of Pattern-Sensitive Epilepsy"). TMS studies demonstrate reduced normal inhibitory influence from visual to motor cortex in PPR-positive patients, supporting a propagation/inhibitory-containment mechanism rather than a purely retinal one (Strigaro et al.). Source-localization studies show occipital-to-parietal-to-frontal recruitment, with thalamic participation becoming more prominent in some generalized events ("Representation and Propagation of Epileptic Activity").

CHD2 supplies a useful genetic benchmark. Galizia and colleagues found rare CHD2 variants enriched in photosensitive epilepsy and showed that *chd2* knockdown enhanced photosensitivity in zebrafish (Galizia et al.). No locus within the present 17p13.3 interval currently has comparable combined human-genetic and controlled-photic experimental evidence.

Our Findings

8. Phase 4A - Dosage Sensitivity and Photosensitivity-Centered Gene Audit

Phase 4A audited all thirteen affected protein-coding genes under a five-level evidence

scale: A for direct heterozygous functional evidence; B for direct heterozygous molecular or biochemical dosage evidence; C for strong perturbational evidence principally from severe, biallelic, knockout, or otherwise non-equivalent models; D for mechanistic plausibility without sufficiently relevant dosage perturbation; and E for insufficient meaningful evidence. Plus and minus modifiers indicate position near a tier boundary. The grade measures dosage/perturbation evidence, not photosensitivity relevance itself.

Gene

Core Phase 4A

Grade

Photosensitivity

Principal limitation

mechanism

priority

INPP5K

PI(4,5)P2/PIP3 5-

B+

High

One-copy physiology

phosphatase; membrane

exists outside retina;

and lysosomal
retinal heterozygous PI
phosphoinositide
phenotype untested.
control PITPNA
PI/PC transfer and local
B-/C+
High
Strong pathway biology
phosphatidylinositol
but PITPNB
substrate availability
compensation and weak
retinal heterozygous
evidence. WDR81
VPS34/PI3P regulation
C+
Moderate-high
Strong severe-loss
and endosome
retinal/developmental
maturation
phenotype;

heterozygous retinal

phenotype absent. RILP

Rab7 effector;

C

Moderate

Strong mechanistic

dynein/dynactin late-

transport evidence;

endosome transport

clean RILP+/- visual

phenotype absent. SERPINF1

PEDF trophic/lipid

B+

Very high

Human heterozygous

signaling through

dosage reduction

PEDF-R/PNPLA2

established; local retinal

functional dose-

response incomplete. RTN4RL1

GPI-anchored neuronal

C+

Moderate
Human KO alters
receptor;
networks, but
neurite/synapse
heterozygous visual-
organization
circuit physiology
untested.

Gene

Core Phase 4A

Grade

Photosensitivity

Principal limitation

mechanism

priority

PRPF8
Core U5 spliceosomal
B-/C+
Very high
Dominant retinal
scaffold and retinal
variants strongly

transcript processing

relevant; whole-gene

haploinsufficiency

unresolved. RPA1

ssDNA protection,

A-

High for stress reserve

Direct human

ATR checkpoint, DNA

haploinsufficiency with

repair

functional defect and

rescue; not retina-

specific. TLCD2

PE/PC fatty-acyl

C+

Very high mechanistic

Strong lipid-

composition and

composition effect;

membrane physical

retinal heterozygous

chemistry

direction unknown. SLC43A2

Neutral essential

C-/D+

Low-moderate

Systemic severe-loss

amino-acid transport

effects; retinal substrate

limitation unproven. SCARF1

C1q-associated

C-/D+

Low-moderate

Clear clearance biology

efferocytosis

but not canonical POS

receptor or primary

photon-response

pathway. SERPINF2

Alpha2-antiplasmin;

B+

Low

Strong biochemical

fibrinolysis inhibition

dosage control with

little photosensitivity

proximity. SMYD4

SAM-dependent

D+

Low

Developmental

methyltransferase;

epigenetic biology but

chromatin regulation

no relevant human

retinal HI phenotype.

The Phase 4A prioritization placed TLCD2, PRPF8, SERPINF1, INPP5K, and PITPNA closest to the chemistry required to create, maintain, or regulate retinal light responses. RPA1, WDR81, RILP, and RTN4RL1 were retained as major stress-, maintenance-, or neural-reserve candidates. SCARF1, SLC43A2, SERPINF2, and SMYD4 remained lower-priority modifiers. This ranking did not imply that the highest-priority genes were proven causal; it identified where the shortest plausible path to the phenotype existed.

The most important conceptual result of 4A was the separation of sensory hypersensitivity from photochemical vulnerability. Sensory hypersensitivity requires the same photons to produce a larger physiological or perceptual response. Photochemical vulnerability requires the same photons to produce more molecular injury. RPA1 fits the second category far better than the first. TLCD2 and PRPF8 were retained as more proximal candidates to

phototransduction or photoreceptor membrane behavior, while SERPINF1, INPP5K, PITPNA, WDR81, and RILP primarily supported maintenance and reserve.

9. Phase 4B - Molecular Perturbation, Biochemical Flux, and Chemical Consequences

Phase 4B strengthened the evidentiary standard: gene function alone was no longer

sufficient. Each mechanism was required to identify a perturbation, a measurable molecular endpoint, direction, magnitude when available, cellular compartment, rescue evidence, and proximity to the visual phenotype. This converted the project from an annotation exercise into a perturbational chemistry audit.

9.1 TLCD2

TLCD2 knockdown changes incorporation of selected long-chain polyunsaturated fatty acids into PC and PE and changes membrane fluidity under palmitate challenge. The key inference was not "TLCD2 loss makes membranes rigid." In the primary experiment, reduction could protect against saturated-fat-induced rigidification. The correct Phase 4B conclusion was that TLCD2 dosage can change membrane composition and therefore potentially alter the highly specialized physical environment of photoreceptor outer-segment membranes, but the direction in a DHA-rich human retina remains unpredictable.

9.2 SERPINF1/PEDF and PNPLA2

The SERPINF1 arm produced one of the strongest molecular chains in the interval: human heterozygous SERPINF1 loss lowers PEDF; PEDF stimulates PEDF-R/PNPLA2-associated phospholipase/lipase biology; PNPLA2 perturbation alters retinal phospholipid species, outer-segment processing, retinal proteins, visual-cycle retinoid mobilization, and ERG physiology. PEDF/DHA signaling additionally connects to neuroprotectin D1 and oxidative stress protection. The project therefore retained SERPINF1 as a

strong RPE/photoreceptor support candidate while explicitly refusing to equate *Pnpla2* heterozygosity with *Serpinf1* heterozygosity.

9.3 PITPNA and INPP5K

PITPNA and INPP5K were found to occupy complementary but nonidentical phosphoinositide roles. PITPNA supports local phosphatidylinositol/phosphatidylcholine transfer and PI availability. INPP5K directly removes the 5-phosphate from PI(4,5)P₂ and PIP₃, with strong-loss experiments demonstrating altered autolysosomal PI(4,5)P₂/PI4P, persistent clathrin-associated tubules, and impaired lysosome regeneration. Phase 4B emphasized that a steady-state lipid concentration could appear normal even if turnover flux falls. Therefore a combined PITPNA/INPP5K dosage defect could theoretically reduce phosphoinositide regulatory reserve without producing a simple monotonic change in total PIP₂.

9.4 WDR81 and RILP

WDR81 and RILP form a sequential cellular-material-flow architecture rather than a single biochemical pathway. WDR81 regulates early-endosomal PI3P and early-to-late maturation; RILP couples Rab7-positive late endosomes to dynein-driven movement. The critical distinction is that endosomal identity, physical transport, terminal degradation, and lysosome recycling are separate control points. Phase 4B therefore described WDR81 → RILP → INPP5K as functional adjacency, not as a proven molecular complex or epistatic chain.

9.5 PRPF8

PRPF8 was treated as a production-side mechanism for photoreceptor machinery. Rather than claiming that PRPF8 is itself phototransduction chemistry, Phase 4B positioned it upstream of the transcript and protein products required for photoreceptor cilia, outer segments, and

signaling. Human retinal models of pathogenic PRPF8 variants strongly support retina-specific splicing vulnerability; however, deletion-level haploinsufficiency remains unresolved.

9.6 RPA1

RPA1 remained the strongest direct heterozygous molecular evidence in the entire interval. Its Phase 4B role was narrowed to stress reserve: reduced RPA1 lowers capacity to manage replication-associated and UV-associated damage in human cells. Independent RPE studies show that blue-light photochemistry can generate oxidative DNA lesions. The project therefore proposed a reserve model in which light-associated lesion production increases demand while RPA1 haploinsufficiency lowers repair/checkpoint capacity. This bridge remains inferential because no retinal RPA1[±] light-challenge experiment exists.

9.7 Lower-Priority Systems

SCARF1 was retained as a post-injury clearance mechanism, not a canonical photoreceptor-outer-segment receptor. SLC43A2 was retained for possible methionine/SAM, glutathione, leucine/mTOR, and other metabolic reserve routes, but no retinal substrate limitation was established. SMYD4 was retained as an epigenetic reserve candidate without retinal targets. SERPINF2 remained a biochemical positive control with little direct photosensitivity relevance. RTN4RL1 was restored to the Phase 4B analysis after an initial omission and was placed in a central-neural branch involving neurite, synaptic, and network organization rather than retinal chemistry.

System**Measured molecular****Observed perturbational****Phase 4B relevance****variable****direction**

TLCD2

PC/PE acyl composition;

Composition changes;

Very high mechanistic,

membrane fluidity

protective under palmitate in

uncertain sign

tested cells SERPINF1/PEDF

PEDF abundance;

Heterozygous PEDF down;

High

PNPLA2/PEDF-R lipid

downstream lipid/retinoid

signaling

effects PITPNA

Local PI/PC transfer and PI

Partial reduction alters lipid

High-moderate

flux

metabolism; developmental

buffering

System

Measured molecular

Observed perturbational

Phase 4B relevance

variable

direction

INPP5K

PI(4,5)P₂ → PI4P; lysosome

PIP₂ up, PI4P down in strong

High

reformation

loss; rescue demonstrated WDR81

VPS34/PI3P early-endosome

PI3P-rich enlarged early

Moderate-high

regulation

endosomes; delayed

maturation RILP

Rab7/dynein late-endosome

Reduced transport and
Moderate
transport
dendritic cargo clearance PRPF8
Retinal transcript splicing
Pathogenic variants alter
Very high
retinal/ciliary splicing RPA1
ATR/DNA-damage response
Direct human HI defect with
Very high for vulnerability
rescue RTN4RL1
Synapse/arborization/network
KO remodels human neuronal
Moderate neural branch
organization
networks SCARF1
Apoptotic-cell clearance
Loss impairs efferocytosis
Moderate-low SLC43A2
Essential-AA transport
Severe-loss systemic transport
Low-moderate

defects SMYD4

Chromatin methylation/HDAC

Developmental epigenetic

Low

interaction

perturbation SERPINF2

Fibrinolytic inhibition

Strong heterozygous

Very low visual

biochemical dosage

10. Phase 4C - Cellular Phenotype and Stress-Response Analysis

Phase 4C asked what the molecular abnormalities actually make retinal or neural cells do

differently. Each claim was graded by genetic state, cell type, phenotype, baseline versus stress dependence, rescue, and proximity to either sensory hypersensitivity or photochemical vulnerability.

10.1 Photoreceptor Integrity

The strongest cellular convergence was photoreceptor integrity. Pathogenic PRPF8 variants produce human retinal-organoid ciliary, outer-segment, and photoreceptor abnormalities. Severe WDR81 dysfunction produces progressive photoreceptor death that is rescued by wild-type WDR81. The SERPINF1-associated PEDF-R/PNPLA2 pathway produces photoreceptor deformation, apoptosis, loss of retinal proteins, and measurable ERG impairment. These mechanisms are independent, so their shared endpoint constitutes phenotypic convergence rather than a single common upstream pathway.

The direction of the physiology mattered. The strongest partial-dose retinal experiment heterozygous *Pnpla2* - showed decreased dark-adapted retinal output, not increased gain. Complete PNPLA2 loss also delayed recovery after photobleaching. Therefore Phase 4C materially weakened a simple "overresponsive retina" hypothesis.

Quantitatively, heterozygous *Pnpla2* mice had retinal thickness of approximately 177.9 $\pm$ 2.156 micrometers at three months versus 209.2 $\pm$ 9.12 micrometers in wild type ($P < .0075$), and 175.5 $\pm$ 5.31 micrometers at seven months versus 203.7 $\pm$ 4.60 micrometers in wild type ($P < .007$). At seven months, the dark-adapted b-wave averaged approximately 215.5 $\pm$ 16.5 microvolts in heterozygotes versus 279.1 $\pm$ 8.91 microvolts in wild type ($P = .0023$). Photopic light-adapted ERG responses were not significantly different at the tested ages. Heterozygous retinas also showed reduced rhodopsin and opsin abundance and substantially reduced overlap between rod-bipolar and synaptic markers. These measurements reinforce that partial PEDF-R/PNPLA2 pathway reduction can impair retinal structure and scotopic physiology without demonstrating increased light-response gain (Bernardo-Colón et al.).

10.2 RPE Support and Outer-Segment Processing

SERPINF1/PEDF became the experimentally strongest RPE arm. *Serpinf1*-null RPE develops senescence-like changes and accumulates rhodopsin and lipids; one-copy loss measurably reduces downstream *Pnpla2* transcript abundance. PNPLA2 deficiency independently impairs retinyl-ester mobilization and visual-cycle recovery. PRPF8 pathogenic variants can also disturb RPE polarity and ciliary biology. By contrast, the Phase 3 RPE spatial convergence of INPP5K + PITPNA + SERPINF1 was only partly validated at the cellular level: direct heterozygous RPE phenotypes for INPP5K and PITPNA remain absent.

10.3 Endosomal/Lysosomal Cellular Phenotype

WDR81 deficiency enlarges early endosomes and delays cargo progression; in the neural-progenitor analysis, mean EEA1-positive early-endosome size increased by approximately 63 percent under WDR81 deficiency. RILP-pathway disruption reduces dendritic late-endosome movement and alters neuronal morphology. INPP5K loss prevents efficient autolysosomal membrane recycling and lysosome replenishment. Together these data support a serial material-handling architecture. They do not prove that heterozygous reduction of all three genes in the same retinal cell produces an additive phenotype.

10.4 Stress Challenge

The stress branch produced a coherent but incomplete protection-repair-clearance model. PEDF/NPD1 biology can protect RPE from oxidative stress; RPA1 haploinsufficiency reduces DNA-damage-response reserve; SCARF1 participates after cell death by clearing apoptotic material. The critical full-CNV experiment - identical visible-light challenge in wild-type versus heterozygous deletion retinal models - has not been performed.

10.5 Central Neural Phenotype

RTN4RL1 became the strongest direct human-neuronal cellular candidate. Complete loss increases neurite growth but decreases synaptic puncta by approximately 30-40 percent; synchronous calcium peaks fall by approximately 40-80 percent and peak amplitudes by approximately 50-70 percent across clones. RILP and WDR81 independently alter dendritic or progenitor phenotypes. MIR22 complete loss worsens later spontaneous epilepsy after status epilepticus: knockout mice averaged roughly 16 seizures per day versus 2.47 in wild type, with mean seizure duration approximately 18.4 seconds versus 11.9 seconds. Chronic heterozygous animals were not followed in the decisive seizure phase, and no photic stimulation was used. The

combined interpretation was altered neural reserve and network construction, not demonstrated hyperexcitability (Wang et al.; Silva et al.).

Locus/system

Model

Genetic state

Cellular phenotype

Phase 4C interpretation

PRPF8

Human retinal organoid

Heterozygous pathogenic

Photoreceptor/ciliary/

High; genotype mismatch

variant

outer-segment defect WDR81

Mouse retina

Severe/homozygous

Photoreceptor death;

High-moderate;

rescue

heterozygous negative SERPINF1/PEDF

Mouse RPE / pathway

+/- and -/-; downstream

RPE processing, retinal

High-moderate

models

PNPLA2

support PNPLA2/PEDF-R

Mouse retina/RPE

+/- and -/-

ERG reduction, retinal

High pathway evidence;

thinning, delayed

gene not deleted

adaptation RPA1

Human LBLs

Heterozygous genomic

Stress-response/

Very high dosage

loss

checkpoint failure; rescue

evidence; nonretinal RTN4RL1

Human neurons

Complete KO

Neurite overgrowth,

High cellular relevance;

synapse loss, lower

not HI

synchronous Ca²⁺ RILP

Cultured neurons

Rab7-RILP disruption

Reduced late-endosome

Moderate

transport, dendrite

defects INPP5K

Myoblast/muscle

Strong loss/KD

Autolysosome

Moderate; nonretinal

enlargement, lysosome

depletion TLCD2

HEK293

siRNA

Membrane-composition

Uncertain direction

change; protection under

palmitate MIR22

Mouse brain

Complete deletion after

Accelerated/worsened

High seizure relevance;

SE

chronic epilepsy

not heterozygous or

photic

11. Phase 4D - Developmental Perturbation and Critical Windows

Phase 4D tested whether affected genes change how retinal and neural systems are built,

rather than only how mature cells function. The key result was distributed timing. No single developmental "hit" explained the interval. Instead, different loci act at progenitor, differentiation, neurite, synaptic, retinal-maturation, and post-insult remodeling stages.

WDR81 was the strongest early corticogenesis candidate. Severe loss reduces mid-to-late radial-glial proliferation and later-born cortical-neuron production/positioning through altered endosomal growth-factor signaling. PITPNA-associated developmental biology operates in a similar broad window but is strongly buffered by PITPNB, preventing transfer of the severe combined-loss phenotype to simple PITPNA heterozygosity.

INPP5K reduction affects neuronal process elaboration and strong-loss ocular development. RTN4RL1 and RILP act later on axon/dendrite and synapse construction, with partly opposing morphological directions. SERPINF1/PEDF contributes to postnatal retinal maturation, vascular development, trophic support, and stress resilience. PRPF8 pathogenic organoids show relatively intact early neural-retina specification followed by late photoreceptor maturation failure. MIR22 has little evidence for gross constitutive brain malformation; its strongest role appears during epileptogenic remodeling after injury.

The developmental phase also strengthened the concept of compensation. PITPNA/PITPNB redundancy, PEDF developmental catch-up in some vascular measures, preserved early PRPF8 retinal specification, and normal baseline MIR22-null anatomy all demonstrate that normal baseline structure does not imply normal reserve. Phase 4D therefore favored a distributed critical-window vulnerability model over a congenital V1-specific or magnocellular lesion.

The noncoding loci were also constrained explicitly. MIR22 is embedded within MIR22HG, so MIR22HG and MIR22 cannot be counted as two independent deletion events supporting the same mechanism. MIR22HG has no independently validated visualdevelopmental mechanism sufficient to add separate weight. PITPNA-AS1 remains dominated by context-dependent cancer-cell literature rather than normal retinal or neural dosage biology. RN7SL105P has no validated locus-specific function that warrants mechanistic weight. These negative assignments are retained in the frozen report to prevent artificial inflation of the multigene convergence model.

Locus

Critical window

Best developmental

4D grade

Principal limitation

endpoint

WDR81

Mid/late corticogenesis

Radial-glial

C+

Severe loss dominates

proliferation;

EGFR/MAPK; late-

born neuron positioning

Locus

Critical window

Best developmental

4D grade

Principal limitation

endpoint

RTN4RL1

Neurite/synapse

Axon/dendrite growth;

C+

Complete KO, not HI

maturation

synapse formation;

network activity SERPINF1

Postnatal retinal

Retinal survival,

C+

Null-heavy evidence

maturation

vascular trajectory,

stress sensitivity PRPF8

Late retinal maturation

Outer-segment and

B-/C+

Pathogenic variant !=

photoreceptor

deletion

specialization INPP5K

Neuronal

Neurite length; ocular

C+

HI unknown

differentiation/eye

morphogenesis under

development

strong loss RILP

Dendritic maturation

Late-endosome

C

Not direct RILP+/-

transport supports

arborization PITPNA

Corticogenesis

Golgi PI signaling,

C-/D+

PITPNB redundancy

polarity, nuclear

migration RPA1

Proliferative reserve

Genome stability and

D+

No direct neural HI

repair

developmental

phenotype MIR22

Post-insult remodeling

Epileptogenic plasticity

E baseline / C+

Chronic heterozygous

after status epilepticus

remodeling

effect untested

12. Phase 4E - Multi-Gene Functional Convergence

Phase 4E tested whether the independently supported effects actually meet. Five levels of

convergence were distinguished: spatial co-localization, molecular-system convergence, sequential-process convergence, phenotypic convergence, and direct combinatorial convergence. The first four were observed to varying degrees; the fifth was not. No relevant primary study jointly reduced two loci from this exact interval and measured a retinal or photosensitivity-related outcome. Accordingly, Phase 4E used "functional convergence" and explicitly withheld "synergy" and "epistasis."

12.1 Membrane Identity and Material Flow

PITPNA, WDR81, and INPP5K converge on phosphoinositide-dependent membrane identity but regulate different lipid pools in different compartments. PITPNA supports PI/PC transfer and PI4P-generating signaling, WDR81 restrains VPS34-dependent PI3P on endosomes, and INPP5K turns over PI(4,5)P₂ on autolysosomal membranes. RILP then links late-endosomal identity to physical transport. This architecture is distributed, not a linear PI → PI3P → PIP₂ cascade.

The most coherent serial cellular architecture was WDR81 → RILP → INPP5K, representing endosome maturation → late-endosome transport → autolysosomal/lysosomal recycling. A serial system can be limited by its weakest stage, so multi-gene effects need not be additive. One defect can mask another, multiple partial defects can remain silent until cargo demand rises, or the system can display threshold behavior.

12.2 Photoreceptor Endpoint Convergence

PRPF8, WDR81, and the SERPINF1/PEDF-R system independently converge on photoreceptor structural and functional integrity through RNA processing, intracellular

maintenance, and trophic/lipid support. TLCD2 adds a plausible membrane-composition route without direct retinal cellular validation. This was the strongest phenotypic convergence identified in Phase 4.

12.3 Stress-Defense Convergence

SERPINF1, RPA1, and SCARF1 map onto different layers of stress response: damage tolerance/protection, DNA-damage response/repair reserve, and post-injury clearance. The architecture is coherent because all three affect how tissue withstands or resolves injury, but they have not been jointly tested in one retinal model.

12.4 Circuit-Construction Convergence

WDR81, INPP5K, RTN4RL1, and RILP converge topologically on neural development and network construction. Directional coherence is weaker. RTN4RL1 knockout increases neurite branching while RILP-pathway disruption reduces dendritic arborization; PITPNA is buffered by PITPNB; and RTN4RL1 knockout decreases synchronous activity. Phase 4E therefore concluded that the interval perturbs multiple controls of network topology without demonstrating a unified hyperexcitable direction.

Convergence module

Principal loci

Topological

Directional coherence

4E grade

convergence

Photoreceptor integrity

PRPF8, WDR81,

Very strong endpoint

Mostly impairment

B

SERPINF1 pathway,

convergence

TLCD2 candidate Endosomal/lysosomal

WDR81, RILP, INPP5K

Very strong sequential

Compatible but not scalar

B-/C+ material flow

convergence RPE outer-segment

SERPINF1;

Strong

Likely impaired

C+ processing

WDR81/RILP/INPP5K

anatomical/mechanistic

candidates Phosphoinositide

PITPNA, WDR81,

Strong molecular-system

Mixed compartment-

C+ membrane identity

INPP5K

specific directions Membrane lipid

TLCD2,

Moderate-strong

Mixed/unknown

C+ composition

SERPINF1/PEDF-R,

PITPNA Photochemical stress

SERPINF1, RPA1,

Strong conceptual

Coherent toward lower

C+/Bdefense

SCARF1

layered reserve

reserve Developmental circuit

WDR81, INPP5K,

Very strong topological

Mixed/opposing

Btopology

RTN4RL1, RILP,

PITPNA Direct retinal sensory

Candidate retinal genes

Weak

Not demonstrated

D/E amplification Epileptogenic reserve

MIR22 plus network-

Moderate opportunity

No consistent

D+/C-

development genes

hyperexcitable direction

13. Phase 4F - Visual-System Relevance and Signal Propagation

Phase 4F mapped the evidence node by node from photoreceptor to photoparoxysmal

response. The major result was asymmetry: the deletion has its strongest biological evidence before the optic nerve and in general network-development systems, while the most phenotypespecific stages - V1 flicker physiology and distributed PPR propagation - remain untested.

13.1 Photoreceptor and RPE Nodes

PRPF8, SERPINF1/PEDF, WDR81, TLCD2, PITPNA, and INPP5K provide meaningful retinal mechanisms. The evidence overwhelmingly concerns structural integrity, adaptation, trafficking, lipid support, or stress tolerance rather than increased photon gain. The strongest partial-dose physiology shows reduced ERG response; PNPLA2 loss slows dark adaptation. Therefore primary retinal amplification was downgraded to weak/unsupported, while altered retinal temporal coding remained plausible but untested.

13.2 RGC, Optic Nerve, and White Matter

PEDF directly supports RGC survival in injury models. RTN4RL1 is expressed in retinal ganglion cells and participates in growth-inhibitory/plasticity systems, but Nogo-receptor family

redundancy is substantial: even combined family loss can leave baseline retinal stratification, optic-nerve myelination, and central RGC projections grossly normal. This weakens a simple optic-pathway miswiring hypothesis.

Human overlapping-CNV evidence nevertheless introduces a credible white-matter branch. The Emrick series demonstrates multifocal leukoencephalopathy in small interstitial 17p13.3 deletions excluding YWHAE and PAFAH1B1, with RTN4RL1 proposed as a candidate. White-matter abnormalities could affect conduction timing and long-range synchronization, but the reported lesions were not shown to selectively involve optic radiations or occipitofrontal pathways and no photic EEG phenotype was reported.

13.3 LGN and V1

Earlier expression analysis found no affected interval gene among the principal defining markers of human magnocellular, parvocellular, or koniocellular dLGN populations. Phase 4 therefore did not support a selective magnocellular mechanism. No deletion-specific LGN hyperexcitability, temporal gain, or thalamocortical entrainment phenotype has been demonstrated.

V1 is the central evidence gap. Human PSE physiology strongly implicates occipital cortex, contrast/gain control, temporal entrainment, inhibition, and network connectivity. The deletion contains cortical and developmental genes, but none has been shown in the heterozygous CNV state to produce abnormal V1 responses to flicker. There is no direct evidence of altered P100, SSVEP, gamma synchronization, alpha stabilization, or V1 inhibitory containment attributable to this CNV.

13.4 Distributed Propagation and Seizure Threshold

PSE often involves occipital-to-parietal-to-frontal spread and abnormal visuomotor connectivity. RTN4RL1, RILP, WDR81, INPP5K, and white-matter findings make altered network topology biologically plausible. Yet no affected-locus or CNV model has demonstrated pathological propagation during visual stimulation. MIR22 is best positioned downstream as a possible modifier of epileptogenic threshold or remodeling rather than a light-trigger mechanism.

Visual node

PSE importance

Deleted-locus evidence

Observed direction

4F status

Photoreceptor

High

PRPF8, SERPINF1

Mostly impaired

Moderate-strong

pathway, WDR81,

structure/function; no

TLCD2 candidate

gain increase RPE

High support role

SERPINF1 strongest;

Reduced support,

Strong support-system

INPP5K/PITPNA

delayed recovery

relevance

candidates Inner retina

Temporal/contrast

PEDF-R pathway;

Reduced ERG / structural

Moderate

filtering

PRPF8 indirect

changes RGC

Final retinal output

PEDF; PRPF8;

Survival/development

Moderate

RTN4RL1 biology

effects Optic nerve/white matter

Timing/conduction

RTN4RL1; overlapping

Structural vulnerability

Moderate

CNV white matter LGN

Relay/modulator

No specific M/P/K

Unknown

Weak

signature V1

Major PPR initiation

General cortical

No deletion-flicker

Major evidence gap

node

mechanisms only

physiology Parietal/frontal network

PPR propagation

RTN4RL1, RILP,

Mixed circuit effects

Plausible, untested

WDR81, white matter Cortical inhibition

Important PSE

No direct CNV E/I

Unknown

Weak

mechanism

phenotype PPR transition

Defining EEG phenotype

No affected-gene/CNV

Unknown

Absent

model General epileptogenic

Seizure conversion

MIR22 severe loss

Lower reserve after insult

Moderate conceptual reserve Clinical photic seizure

Final phenotype

No exact-CNV evidence

Unknown

Unresolved

14. Phase 4G - Integrated Adjudication

Phase 4G formally separated biological consequence from phenotype attribution. The

CNV is very likely biologically non-neutral: it includes directly dosage-responsive genes, multiple retinal pathways, strong severe-loss or pathogenic-variant mechanisms, neuraldevelopment genes, and a clinically relevant overlapping-deletion white-matter phenotype. However, biological consequence is not equivalent to epileptic-photosensitivity causation.

Broad 17p13.3 epilepsy literature cannot be transferred wholesale because many published deletion syndromes include PAFAH1B1 and/or YWHAE/CRK, which are outside this interval. The closest human CNV precedent supports white-matter vulnerability rather than a

defined photosensitive-epilepsy syndrome. Likewise, no locus in this interval meets the CHD2 benchmark of human photosensitivity association plus controlled photic experimental validation.

The integrated causal ladder therefore stops before the disease-specific physiological steps. Genomic lesion and dosage alteration are established. Molecular consequences are established for selected genes. Relevant cellular phenotypes are established in constituent-gene models. Multi-gene functional convergence is supported. A full-CNV retinal phenotype is not established. Abnormal visually evoked cortical physiology is not established. A PPR and a photosensitive seizure caused by the exact CNV are not established.

Claim

Phase 4G adjudication

CNV biologically non-neutral

Supported Molecular reserve altered

Strongly supported for selected loci Retinal mechanisms present

Strongly supported Neurodevelopmental mechanisms present

Strongly supported Multi-gene functional convergence

Supported Multi-gene synergy/epistasis

Not demonstrated Retinal photon-response amplification

Unsupported Retinal light-injury vulnerability

Plausible; not directly demonstrated Abnormal retinal temporal coding

Open hypothesis Long-range network modification

Plausible; partially supported by overlapping CNVs V1 hyperexcitability from CNV

Not demonstrated Abnormal visually driven cortical propagation

Plausible; not demonstrated General epileptogenic reserve reduction

Plausible; strongest evidence through MIR22 complete loss Photoparoxysmal response caused by exact CNV

Not demonstrated Epileptic photosensitivity caused by exact CNV

Insufficient evidence / unresolved Best current classification

Mechanistically plausible candidate modifier of visual-neural and epileptogenic reserve

15. Frozen Phase 4 Mechanistic Model

The strongest integrated model after 4A-4G is a multi-layer visual-neural functionalreserve lesion. It does not require every deleted gene to be individually haploinsufficient, nor does it require every perturbation to move in the same direction. Instead, the interval can lower or alter reserve at several control layers while paralogs, developmental compensation, homeostatic adaptation, and opposite-direction effects preserve baseline function. 17p13.3 deletion -> retinal maintenance reserve down + membrane/endosomal reserve down + stress-response reserve down + developmental/network organization altered

Under this model, baseline demand can remain below the reduced reserve of the heterozygous system, allowing apparently normal anatomy or physiology. Increased demand intense or repetitive visual stimulation, oxidative stress, developmental maturation, metabolic stress, illness, or a separate epileptogenic influence - may expose deficits that are otherwise compensated.

If the CNV ultimately contributes to epileptic photosensitivity, Phase 4 indicates that the most plausible route is not excessive photon capture. The stronger hypothesis is altered retinal timing/adaptation and/or altered circuit construction or white-matter/network organization, combined with reduced epileptogenic reserve, producing a lower threshold for abnormal

propagation of patterned visual activity. This remains a testable hypothesis rather than a demonstrated mechanism.

16. Hypotheses Retained, Downgraded, or Rejected

16.1 Retained as Best-Supported

- The deletion is biologically consequential and can reduce molecular/cellular reserve.
- Retinal and RPE maintenance pathways are genuinely represented in the interval.
- Photoreceptor structural integrity is the strongest phenotype-level convergence.
- Endosomal/lysosomal material handling is the strongest serial mechanistic convergence.
- Distributed developmental and network-organization effects are plausible and experimentally supported under strong perturbation.
- The deletion is a plausible candidate modifier of visual-neural and epileptogenic reserve.

16.2 Retained but Unproven

- Increased retinal vulnerability to the same light dose.
- Abnormal retinal temporal coding or adaptation at provocative flicker frequencies.
- Altered long-range visual-network propagation because of circuit-development or white-matter changes.
- A modest lowering of epileptogenic threshold in the heterozygous CNV state.

16.3 Downgraded or Unsupported

- A simple intrinsically hyperresponsive retina.
- A selective magnocellular lesion.
- A deletion-wide V1-specific developmental program.
- Generalized RTN4RL1-driven hyperexcitability.
- Uniformly harmful TLCD2-mediated membrane rigidification.

- Proven synergy or epistasis among deleted genes.
- Established direct causation of PPR or photosensitive seizures by this exact CNV.

17. Definitive Experiments Implied by Phase 4

The single highest-value experimental strategy is an isogenic reconstruction of the actual heterozygous deletion rather than a collection of unrelated null models. Wild-type, full-CNV heterozygous, corrected/rescued, and selected single-gene heterozygous lines should be differentiated into retinal organoids, RPE, and connected neural systems. The engineered interval should reproduce the reported hg19 coordinates as closely as technically possible, particularly the boundary structures of INPP5K and RTN4RL1.

The retinal arm should measure photoreceptor structure, outer-segment formation, retinal/RPE lipidomics, phosphoinositide pools, outer-segment phagocytosis and degradation, visual-cycle retinoids, dark adaptation, ERG-like responses, oxidative/DNA damage, and recovery after calibrated light exposure. The neural arm should measure progenitor behavior, neurite morphology, synapse density, E/I balance, calcium dynamics, MEA firing and bursting, frequency entrainment, phase locking, and long-range propagation.

For photosensitivity specifically, patterned visual stimulation must be treated as a transfer-function problem. Identical stimuli should be delivered across relevant flicker frequencies, contrasts, wavelengths, and spatial structures while measuring the transformation from retinal output through thalamic and V1-like networks into distributed activity. Only demonstration of genotype-dependent pathological propagation, followed by correction or rescue, would close the major causal gap identified in Phase 4.

Summary

Phase 4 converts the 17p13.3 interval from a list of deleted genes into a disciplined

mechanistic model. The interval contains several genes with real dosage or perturbational consequences, and those consequences repeatedly touch retinal support, photoreceptor integrity, membrane identity, intracellular trafficking, DNA-damage response, neuronal development, synaptic organization, and network plasticity. The biological signal is therefore nonrandom and medically relevant.

At the same time, the phase narrows rather than broadens the photosensitivity claim. The data do not show that the retina is intrinsically overresponsive to individual photons. The best retinal physiology instead shows reduced response or delayed recovery. The best centralneuronal RTN4RL1 model shows network remodeling with reduced synchronous activity. The strongest seizure-related locus, MIR22, modifies post-insult epileptogenesis under complete loss but has not been shown to create a photic phenotype in heterozygotes.

The strongest surviving interpretation is reduced or altered functional reserve across multiple layers. Photoreceptor/RPE systems may be less resilient; endosomal and lysosomal material handling may have less capacity; damage-response reserve may be reduced; and circuit construction or long-range communication may be subtly altered. Compensation and redundancy can keep baseline function apparently normal. The model predicts that differences may emerge under increased demand or repeated challenge rather than as an obligatory congenital deficit.

For epileptic photosensitivity, the unresolved stage is now precisely located. Human disease physiology requires the transformation of patterned visual input into abnormal occipital and distributed cortical activity. The present CNV has not been shown to alter V1 flicker physiology, cortical inhibition, 10-20-Hz entrainment, long-range visual-network propagation, a

photoparoxysmal response, or visually triggered seizures. Therefore the exact CNV cannot currently be classified as an established cause of epileptic photosensitivity.

The frozen Phase 4 classification is: biologically consequential CNV; high retinal/visual mechanistic relevance; moderate-to-high support for reduced functional reserve; low support for direct sensory amplification; moderate support for developmental/network modification; limited-to-moderate evidence for general epileptogenic modification; insufficient evidence for abnormal visually driven cortical propagation; and insufficient evidence for direct epileptic photosensitivity causation. The best current label is "mechanistically plausible candidate modifier of visual-neural and epileptogenic reserve."

Vocabulary

Adaptation. Adjustment of retinal or neural responsiveness during or after sustained/repeated

stimulation; dark adaptation specifically refers to recovery of visual sensitivity after photobleaching.

Autolysosomal reformation (ALR). Membrane-recycling process that regenerates lysosomes

from autolysosomes after prolonged autophagy; INPP5K is required for normal ALR.

Biallelic loss. Disruption of both gene copies. Biallelic loss is not equivalent to the heterozygous

one-copy loss produced by this CNV.

Candidate modifier. A factor plausibly capable of changing severity, threshold, penetrance, or

expression of a phenotype without being established as a sufficient primary cause.

Cellular phenotype. A measurable change in cell structure, survival, trafficking, signaling, electrophysiology, or stress response following perturbation.

CNV. Copy-number variant; a genomic deletion or duplication altering the number of copies of one or more loci.

Directional convergence. Multiple perturbations pushing a shared endpoint in the same direction. This is distinct from merely affecting the same system.

Dosage sensitivity. The degree to which changing gene copy number alters transcript, protein, molecular function, cell behavior, or clinical phenotype.

Efferocytosis. Recognition and engulfment of apoptotic cells or cellular debris by phagocytic cells; SCARF1 participates in C1q-associated efferocytosis.

Electroretinography (ERG). Measurement of retinal electrical responses to light. The a-wave primarily reflects photoreceptor activity; the b-wave reflects downstream inner-retinal responses.

Endolysosomal system. The linked endosome-to-lysosome network that sorts, transports, degrades, and recycles cellular cargo.

Epileptogenic reserve. Conceptual resistance of a neural network to transitions into persistent or

recurrent epileptic activity; lower reserve implies easier pathological transition under challenge. Epistasis. A genetic interaction in which the effect of one locus depends on the state of another.

Phase 4 did not demonstrate epistasis among deleted loci.

Functional convergence. Independent genes or mechanisms affecting the same molecular system, cellular process, anatomical compartment, developmental operation, or phenotype endpoint.

Functional reserve. Difference between a system's maximal or available biological capacity and

the physiological demand placed on that system.

Gain. Ratio between change in output and change in input. Increased sensory gain means a larger physiological response to the same stimulus.

Haploinsufficiency. A condition in which one functional gene copy is insufficient for normal molecular, cellular, or clinical function.

Heterozygous loss. Loss or disruption of one of two gene copies. It does not automatically imply

exactly 50 percent protein abundance or function. Hypomorph. An allele that retains partial gene function rather than producing complete loss.

Isogenic. Genetically matched experimental lines differing only at the engineered locus or CNV

of interest, minimizing background-genome confounding.

Koniocellular, magnocellular, parvocellular. Major subdivisions of primate visual-thalamic organization. Phase 4 did not identify selective magnocellular involvement by the deletion.

Lateral geniculate nucleus (LGN). Principal visual relay nucleus of the thalamus between retinal ganglion cells and primary visual cortex.

MIR22. MicroRNA locus within the deletion. Complete loss worsens chronic epilepsy after an

epileptogenic insult in mice; heterozygous photic effects remain unknown.

Neuroprotectin D1 (NPD1). DHA-derived lipid mediator with anti-inflammatory and anti-apoptotic effects in RPE under oxidative stress.

Outer segment. Specialized photoreceptor compartment containing stacked photopigment-rich

membranes where photon absorption and phototransduction begin.

PEDF. Pigment epithelium-derived factor, encoded by SERPINF1; a secreted retinal trophic,

antiangiogenic, lipid-linked, and cytoprotective protein.

Photochemical vulnerability. Greater molecular or cellular injury from the same photon exposure, distinct from sensory hypersensitivity.

Photoparoxysmal response (PPR). Abnormal epileptiform EEG response evoked by intermittent photic or pattern stimulation; it can occur with or without a clinical seizure.

Photosensitive epilepsy (PSE). Epilepsy in which visual stimulation can provoke abnormal EEG activity and, in some patients, clinical seizures.

Phototransduction. Conversion of absorbed photons into an electrical response in

photoreceptors through rhodopsin/opsin, transducin, PDE6, cGMP, and cyclic-nucleotide-gated channel signaling.

Phosphoinositide. Phosphorylated phosphatidylinositol lipid that helps define membrane

identity, trafficking, signaling, and protein recruitment.

PI3P. Phosphatidylinositol 3-phosphate, enriched on early endosomes and produced by VPS34;

WDR81/WDR91 negatively regulate this system.

PI4P. Phosphatidylinositol 4-phosphate, a membrane signaling lipid and product of INPP5K-

mediated PI(4,5)P₂ dephosphorylation.

PI(4,5)P₂. Phosphatidylinositol 4,5-bisphosphate, a membrane signaling lipid whose 5-phosphate is removed by INPP5K.

Retinal ganglion cell (RGC). Output neuron of the retina whose axons form the optic nerve and

project to visual brain targets.

Retinal pigment epithelium (RPE). Support epithelium adjacent to photoreceptor outer

segments; essential for phagocytosis, retinoid cycling, lipid handling, trophic support, and metabolic homeostasis.

Sensory hypersensitivity. An abnormally large perceptual or physiological response to the same

light stimulus.

Spliceosome. Large ribonucleoprotein machinery that removes introns from pre-mRNA;
PRPF8

is a central U5 spliceosomal scaffold.

Steady-state visual evoked potential (SSVEP). Stimulus-frequency-locked cortical response
 produced by periodic visual stimulation; useful for evaluating visual entrainment and
 temporal stability.

Synergy. A combined effect greater than expected under a specified additive model. Phase
 4 did

not demonstrate synergy among deleted genes.

Temporal functional convergence. Different loci influencing the same biological endpoint at
 different developmental or physiological stages rather than acting simultaneously.

Topological convergence. Different perturbations affecting the same biological network or
 structural endpoint even if they push that endpoint in different directions.

V1. Primary visual cortex, the principal cortical recipient of LGN visual input and a major
 initiation node for photoparoxysmal responses.

Visual-neural reserve. Integrated capacity of retinal, visual-pathway, and cortical systems
 to

maintain stable function under physiological or pathological demand.

Reproducible Formulas

The formulas below are separated into biochemical reactions, quantitative definitions, and conceptual models. Biochemical reactions summarize experimentally established chemistry. Quantitative definitions can be applied directly to future datasets. Conceptual equations are explicitly labeled and should not be mistaken for validated physiological laws.

Reproducible Formulas

Formula / definition	Expression	Interpretation
INPP5K 5-phosphatase reaction	$\text{PI(4,5)P}_2 + \text{H}_2\text{O} \rightarrow \text{PI4P} + \text{P}_i$	Established biochemical reaction.
INPP5K PIP3 reaction	$\text{PI(3,4,5)P}_3 + \text{H}_2\text{O} \rightarrow \text{PI(3,4)P}_2 + \text{P}_i$	Established substrate reaction; do not infer in vivo flux from K_m alone.
VPS34 reaction	$\text{PI} + \text{ATP} \rightarrow \text{PI3P} + \text{ADP}$	Established kinase chemistry; WDR81/WDR91 regulate the associated endosomal PI3K system.
PLC branch	$\text{PI(4,5)P}_2 \rightarrow \text{DAG} + \text{IP}_3$	Canonical signaling branch relevant to calcium-regulatory context.
Dynein/ATPase work	$\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{P}_i + \text{mechanical work}$	Represents ATP-powered motor/remodeling work; RILP recruits dynein rather than catalyzing ATP hydrolysis itself.

Reproducible Formulas (continued)

Formula / definition	Expression	Interpretation
PEDF-R/PNPLA2 phospholipase	phospholipid + H ₂ O -> lysophospholipid + free fatty acid	Simplified representation of phospholipase/lipase chemistry.
RPA binding	ssDNA + RPA <-> RPA-coated ssDNA	Binding equilibrium underlying ATR recruitment and DNA-damage response.
Rhodopsin conformational equilibrium	MI <-> MII	Simplified metarhodopsin I/metarhodopsin II equilibrium influenced by membrane environment.
Lipid-radical initiation	LH + radical -> L radical	Simplified initiation of lipid peroxidation.
Lipid-peroxyl propagation	L radical + O ₂ -> LOO radical	Simplified oxygen addition step in lipid peroxidation.

Reproducible Formulas (continued)

Formula / definition	Expression	Interpretation
Methionine activation	Met + ATP → SAM	Simplified methionine-to-S-adenosylmethionine activation route.
SAM-dependent lysine methylation	Lys + SAM → methyl-Lys + SAH	Simplified methyltransferase reaction relevant to SMYD4.
Steady-state flux balance	$dX/dt = J_{in} - J_{out}$	At steady state, $dX/dt = 0$ even when turnover flux is high; a normal concentration does not prove normal flux.
Functional-reserve condition	phenotype risk rises when $D(t) > C_{max}$	Conceptual model: $D(t)$ is physiological demand and C_{max} is available system capacity.
Light-injury function	$I(H)$	I is measured injury at photon/light exposure H . A vulnerability phenotype requires $I_{CNV}(H) > I_{WT}(H)$ for the same H .

Reproducible Formulas (continued)

Formula / definition	Expression	Interpretation
Light-injury threshold	H_{crit} = exposure producing a predefined injury criterion	Direct vulnerability would be supported if $H_{crit,CNV} < H_{crit,WT}$.
Neural response gain	$G = \Delta \text{output} / \Delta \text{input}$	Operational definition. Sensory/neural amplification requires $G_{CNV} > G_{WT}$ under matched input.
Retinal temporal gain	$G_R(f) = R(f) / S(f) $	$R(f)$ is retinal output amplitude at frequency f ; $S(f)$ is stimulus amplitude. Useful for testing flicker-frequency sensitivity.
Serial bottleneck capacity	J_{max} approximately $\min(C1, C2, ..., Cn)$	Conceptual model for sequential endosomal/lysosomal stages; overall throughput can be limited by the weakest stage.
Layered lesion burden	$dL/dt = D - P - R$	Conceptual model: D = damage generation, P = prevention/tolerance, R = repair capacity. Used only as a qualitative reserve representation.

Reproducible Formulas (continued)

Formula / definition	Expression	Interpretation
Factorial interaction model	$Y = \text{beta0} + \text{betaA} * A + \text{betaB} * B + \text{betaAB} * (A * B) + \text{epsilon}$	Reproducible test for non-additive interaction. betaAB tests departure from the specified additive model.
Developmental timing model	$P = f(D_{\text{early}}, D_{\text{late}}, D_{\text{adult}})$	Conceptual decomposition of mature phenotype into early developmental, late maturation, and adult dosage effects.
Photosensitive-network threshold	$G_N * I > \text{Theta}_E$	Conceptual model: I = visual input, G_N = network propagation gain, Theta_E = threshold for pathological synchronization.

Reproducibility rule for future Phase 5 and Phase 6 work: formulas that represent measured quantities must be parameterized from independent data, uncertainty intervals must be propagated, and no conceptual equation should be fitted as if it were a validated physiological law without an explicit calibration dataset.